

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Analog IC design [ECE_4408]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain with neat diagram the punch-through effect in MOSFET and state the condition under which it occurs. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) Explain with neat diagram the channel length modulation in MOSFET and write the expression for I_{Dsat} for nMOS. (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) Design the CS stage of Fig.2A for a voltage gain of 5, an input impedance of 50 kohm, and a power budget of 5 mW. Assume $\mu_n C_{ox} = 100 \mu A/V^2$, $V_{TH} = 0.5V$, $\lambda = 0$, and $V_{DD} = 1.8 V$. Also, assume a voltage drop of 400 mV across R_S . (5)

A)

Design the CS stage of Fig.2A for a voltage gain of 5, an input impedance of 50 kohm, and a power budget of 5 mW. Assume $\mu_n C_{ox} = 100 \mu A/V^2$, $V_{TH} = 0.5V$, $\lambda = 0$, and $V_{DD} = 1.8 V$. Also, assume a voltage drop of 400 mV across R_S .

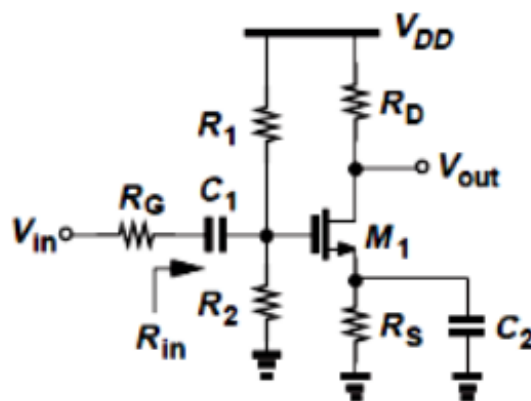


Fig 2A

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) The cascode current source shown in Figure 2B must be designed for a bias current of 0.5 mA. Assume $\mu_n C_{ox} = 100 \mu A/V^2$ and $V_{TH} = 0.4 V$. (5)

(i) Neglecting channel-length modulation, compute the required value of V_{b2} . What is the minimum

tolerable value of V_{b1} if M_2 must remain in saturation?

(ii) Assuming $\lambda = 0.1 \text{ V}^{-1}$, calculate the output impedance of the circuit.

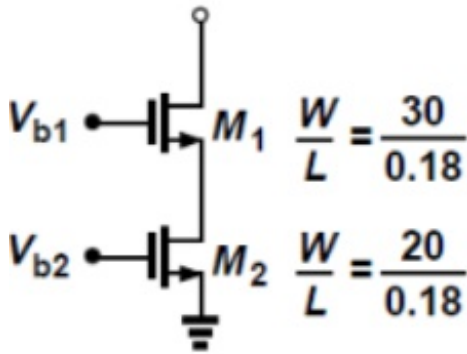


Fig 2B

Course Outcome: "CO2" / Cognitive Level: "Analysis"

3) The identical **MOSFETs** M_1 and M_2 in the circuit shown in (5)

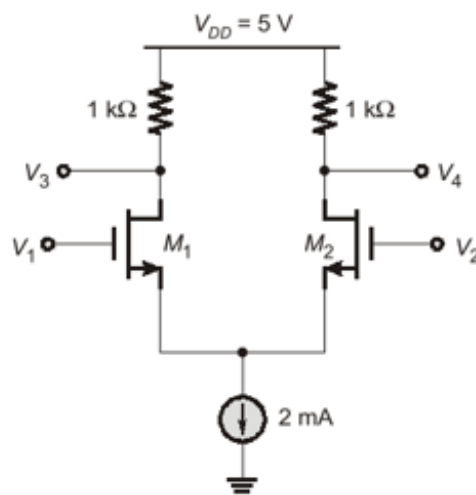
A) Fig 1 B are ideal and biased in the saturation region. M_1 and M_2 each have a transconductance $g_m = 5 \text{ mS}$.

The input signals (in volts) are:

$$V_1 = 2.5 + 0.01 \sin \omega t$$

$$V_2 = 2.5 - 0.01 \sin \omega t$$

The circuit has a tail current source of **2 mA** and two $1 \text{ k}\Omega$ load resistors connected to $V_{DD} = 5 \text{ V}$, as shown. Find the output voltage V_3 .



Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) Explain the working of OTA. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

4) Identify all the capacitances and reduce the circuit shown in Fig 4A to compute the poles. Assume (5)
 $\lambda = 0$.

A)

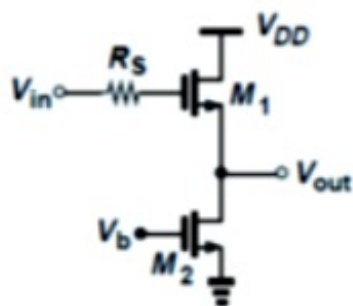


Fig 4A

Course Outcome: "CO4" / Cognitive Level: "Analysis"

B)

Explain the slewing in Op-amps.

(5)

Course Outcome: "CO4" / Cognitive Level: "Application"

5)

Two transistors, Q1 and Q2, operate at the same collector

(5)

A)

current but have emitter area ratio $N = \frac{A_2}{A_1} = 8$.

Temperature $T = 300$ K. Thermal voltage $V_T = \frac{kT}{q}$.

Use $k/q = 8.617333 \times 10^{-5}$ V/K. Resistor

$R = 10$ k Ω placed so that $I_{PTAT} = \frac{\Delta V_{BE}}{R}$.

Find ΔV_{BE} and I_{PTAT} at 300 K.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

B)

Explain the working of a linear voltage regulator.

(5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_4408:Analog IC design

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_4408:Analog IC design

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)

2	Understand	2	10(20%)
3	Application	3	15(30%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)